Head-Orientation Steering on Real Glasses Recordings (EasyCom)

Speech Intelligibility Benchmark Lab, Round 7 (6-ch glasses, restaurant, 6DOF head poses; held-out test sessions 10, 11)

2026-10-24

1. Bottom line

1. **Question.** Does steering a causal ≤ 8 ms beamformer by the wearer's head orientation beat steering it by the round-6 acoustic tracker, on real recordings from glasses in a noisy restaurant?
2. **Yes, clearly.** On the held-out test (434 items with words, sessions 10-11), head-steered superdirective 8 ms gains **+0.066 [+0.029, +0.106]** word-correct over the unprocessed front mic (Holm $p < 0.001$). The tracked version **loses** (−0.064 [−0.090, −0.039]). Head-prior hybrid SD 8 ms: **+0.070 [+0.029, +0.116]**. Oracle reference direction (REF, SD 8 ms): **+0.173 [+0.147, +0.197]**.
3. **But it only ties round-5 delay-and-sum.** Head-steered SD 8 ms vs R5 DS (non-causal, 32 ms): +0.001 [−0.043, +0.050]; hybrid +0.005 [−0.042, +0.057]. So a causal 8 ms filter with a free head prior matches what a 32 ms blind-direction filter gives; it does not beat it. Time-domain DS (0.75-1 ms) with head steering gives nothing (+0.004).
4. **Overlap and wearer-addressed speech.** Overlapped items (n = 248): head SD 8 ms +0.034 [+0.005, +0.066], hybrid +0.038 [+0.005, +0.074], neither survives Holm (p = 0.08-0.11), and both are below R5 DS (−0.044, −0.040). Items addressed to the wearer (n = 85): head SD 8 ms +0.072 [+0.021, +0.136] (Holm p = 0.063), tracked −0.033. The oracle gains +0.164 on overlap and +0.135 to the wearer.
5. **Onsets.** Head azimuth is within 30° of the talker in **54% of 0-0.25 s frames after a turn starts** (71% when addressed to the wearer; 46% on overlapped turns), vs **35%** for the tracker (56% to the wearer). After 4 s the tracker catches up on 10°: 49% vs 27% for head. The ceiling is that the head is only a coarse (±30°) cue: within 10° only 21% at onset.
6. **Steering error drives the gain.** Items steered within 10° gain +0.156 (SD 8 head) over front; 45-90° error loses −0.069. Spearman(error, gain) ≈ −0.37.
7. **Caveat.** ASR is Whisper base.en as a proxy for intelligibility, not human listeners. EasyCom audio is CC-BY-NC and is not in any deliverable. References and DOA are built from poses and close mics, and 20 test items were flagged as poorly aligned (kept in the main table; the table without them is in section 4).

2. What ran

- **Data:** EasyCom (Main + Calibration, 6-ch glasses: ch1-4 frame mics, ch5-6 ear-canal mics), 16 kHz. Tune on val sessions 4/12 (sub-sample of 300 items), test once on sessions 10/11 (472 items; 434 with words).
- **Methods:** hs = head-steered (ATF-interpolated look direction from the 6DOF head pose); trk = round-6 causal SRP tracker adapted to ch1-4; hyb = head prior plus tracker when it is confident (window ±20°, threshold 8); ref = oracle direction from poses; each with DS time-domain, SD 4 ms, SD 8 ms. R5 DS and unprocessed mics as baselines.
- **Frozen on val before test:** front mic = ch2; WNG floor −6 dB; hybrid window ±20°, threshold 8 (results/r7/frozen_r7.json).
- **Scoring:** faster-whisper base.en, word-correct = hits / reference words; cluster bootstrap over turns' recordings, 95% CI; Holm within family. Objective SI-SDR and ESTOI also computed (summary_r7.md).

3. Methods and latencies

Algorithmic latency (excluding compute): time-domain DS 0.75 ms (head) to 1.07 ms (tracked/ref, crossfade); SD 4 ms = 4 ms; SD 8 ms = 8 ms; R5 DS = 32 ms. Measured compute per hop (idle-ish, load ≈ 0.8, one thread):

component	per hop (ms)	p99 (ms)	hop (ms)	CPU fraction
head DS td	0.015	0.016	2	0.007
head SD 4 ms	0.020	0.024	2	0.010
head SD 8 ms	0.022	0.028	4	0.006
tracked SD 8 ms	0.024	0.028	4	0.006
tracker step (ATF)	0.86	0.96	16	0.054
hybrid logic	0.012	n/a	16	0.001

Compute is a non-issue; head steering needs no tracker at all.

4. Test results (word-correct, Δ vs unprocessed front mic and vs R5 DS)

Word-correct, All test items with words (align_suspect kept) (n = 434)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.343			-0.065 [-0.086, -0.043]	
Ear-canal mics mean ch5/6 (reference only)	0.267	-0.076 [-0.110, -0.045]		-0.142 [-0.178, -0.108]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.408	+0.065 [+0.044, +0.085]			
Head-steered DS, time-domain (ATF delays)	0.347	+0.004 [-0.031, +0.036]	1.000	-0.062 [-0.102, -0.021]	0.010
Head-steered superdirective 8 ms	0.409	+0.066 [+0.029, +0.106]	<0.001	+0.001 [-0.043, +0.050]	1.000
Head-steered superdirective 4 ms	0.404	+0.061 [+0.024, +0.102]	0.003	-0.004 [-0.048, +0.042]	1.000
Tracked DS time-domain	0.309	-0.034 [-0.053, -0.015]	<0.001	-0.099 [-0.122, -0.077]	<0.001
Tracked SD 8 ms	0.279	-0.064 [-0.090, -0.039]	<0.001	-0.129 [-0.160, -0.097]	<0.001
Tracked SD 4 ms	0.282	-0.061 [-0.089, -0.036]	<0.001	-0.126 [-0.160, -0.095]	<0.001
Head-prior hybrid DS time-domain	0.340	-0.003 [-0.039, +0.033]	1.000	-0.069 [-0.110, -0.028]	0.006

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Head-prior hybrid SD 8 ms	0.413	+0.070 [+0.029, +0.116]	<0.001	+0.005 [-0.042, +0.057]	1.000
Head-prior hybrid SD 4 ms	0.403	+0.060 [+0.021, +0.098]	<0.001	-0.005 [-0.050, +0.043]	1.000
REF-DOA SD 8 ms (oracle)	0.515	+0.173 [+0.147, +0.197]	<0.001	+0.107 [+0.078, +0.136]	<0.001
REF-DOA DS time-domain (oracle)	0.457	+0.114 [+0.094, +0.132]	<0.001	+0.049 [+0.028, +0.072]	<0.001

Word-correct, Without align_suspect items (n = 414)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.352			-0.068 [-0.088, -0.045]	
Ear-canal mics mean ch5/6 (reference only)	0.273	-0.079 [-0.112, -0.046]		-0.147 [-0.183, -0.113]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.419	+0.068 [+0.045, +0.089]			
Head-steered DS, time-domain (ATF delays)	0.355	+0.003 [-0.032, +0.037]	1.000	-0.064 [-0.106, -0.021]	0.030
Head-steered superdirective 8 ms	0.420	+0.068 [+0.026, +0.109]	0.007	+0.000 [-0.044, +0.048]	1.000
Head-steered superdirective 4 ms	0.416	+0.064 [+0.023, +0.108]	0.008	-0.003 [-0.051, +0.046]	1.000
Tracked DS time-domain	0.318	-0.034 [-0.053, -0.014]	0.007	-0.101 [-0.126, -0.077]	<0.001
Tracked SD 8 ms	0.287	-0.065 [-0.091, -0.039]	<0.001	-0.132 [-0.165, -0.099]	<0.001
Tracked SD 4 ms	0.291	-0.061 [-0.088, -0.034]	<0.001	-0.129 [-0.163, -0.096]	<0.001
Head-prior hybrid DS time-domain	0.349	-0.003 [-0.039, +0.035]	1.000	-0.070 [-0.114, -0.024]	0.030
Head-prior hybrid SD 8 ms	0.424	+0.072 [+0.030, +0.118]	0.007	+0.004 [-0.044, +0.059]	1.000
Head-prior hybrid SD 4 ms	0.415	+0.063 [+0.023, +0.105]	0.008	-0.005 [-0.055, +0.047]	1.000
REF-DOA SD 8 ms (oracle)	0.528	+0.176 [+0.149, +0.202]	<0.001	+0.108 [+0.079, +0.140]	<0.001

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
REF-DOA DS time-domain (oracle)	0.470	+0.118 [+0.098, +0.137]	<0.001	+0.050 [+0.029, +0.071]	<0.001

Word-correct, Overlapped items (n = 248)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.256			-0.078 [-0.098, -0.056]	
Ear-canal mics mean ch5/6 (reference only)	0.217	-0.040 [-0.071, -0.006]		-0.117 [-0.148, -0.085]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.334	+0.078 [+0.055, +0.098]			
Head-steered DS, time-domain (ATF delays)	0.250	-0.007 [-0.036, +0.025]	0.714	-0.084 [-0.110, -0.054]	<0.001
Head-steered superdirective 8 ms	0.290	+0.034 [+0.005, +0.066]	0.084	-0.044 [-0.077, -0.005]	0.064
Head-steered superdirective 4 ms	0.288	+0.032 [+0.006, +0.059]	0.108	-0.046 [-0.078, -0.012]	0.064
Tracked DS time-domain	0.232	-0.024 [-0.052, +0.005]	0.354	-0.102 [-0.125, -0.075]	<0.001
Tracked SD 8 ms	0.216	-0.041 [-0.062, -0.018]	<0.001	-0.119 [-0.143, -0.093]	<0.001
Tracked SD 4 ms	0.219	-0.037 [-0.060, -0.012]	0.024	-0.115 [-0.136, -0.090]	<0.001
Head-prior hybrid DS time-domain	0.239	-0.017 [-0.043, +0.013]	0.460	-0.095 [-0.121, -0.062]	<0.001
Head-prior hybrid SD 8 ms	0.294	+0.038 [+0.005, +0.074]	0.108	-0.040 [-0.076, -0.001]	0.064
Head-prior hybrid SD 4 ms	0.285	+0.028 [+0.001, +0.057]	0.136	-0.049 [-0.084, -0.012]	0.064
REF-DOA SD 8 ms (oracle)	0.421	+0.164 [+0.130, +0.201]	<0.001	+0.087 [+0.054, +0.124]	<0.001
REF-DOA DS time-domain (oracle)	0.369	+0.113 [+0.083, +0.140]	<0.001	+0.035 [+0.013, +0.056]	<0.001

Word-correct, Non-overlapped items (n = 186)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.458			-0.049 [-0.086, -0.011]	
Ear-canal mics mean ch5/6 (reference only)	0.333	-0.126 [-0.171, -0.083]		-0.174 [-0.230, -0.124]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.507	+0.049 [+0.009, +0.083]			
Head-steered DS, time-domain (ATF delays)	0.476	+0.018 [-0.048, +0.072]	1.000	-0.031 [-0.112, +0.038]	0.772
Head-steered superdirective 8 ms	0.569	+0.110 [+0.042, +0.169]	0.010	+0.061 [-0.011, +0.130]	0.714
Head-steered superdirective 4 ms	0.559	+0.101 [+0.021, +0.165]	0.016	+0.052 [-0.024, +0.121]	0.772
Tracked DS time-domain	0.411	-0.047 [-0.075, -0.023]	<0.001	-0.096 [-0.136, -0.061]	<0.001
Tracked SD 8 ms	0.364	-0.094 [-0.134, -0.050]	<0.001	-0.143 [-0.201, -0.078]	<0.001
Tracked SD 4 ms	0.366	-0.092 [-0.132, -0.051]	<0.001	-0.141 [-0.206, -0.080]	<0.001
Head-prior hybrid DS time-domain	0.473	+0.015 [-0.050, +0.079]	1.000	-0.034 [-0.110, +0.041]	0.772
Head-prior hybrid SD 8 ms	0.571	+0.113 [+0.039, +0.177]	0.006	+0.064 [-0.016, +0.138]	0.714
Head-prior hybrid SD 4 ms	0.560	+0.102 [+0.031, +0.164]	0.016	+0.053 [-0.024, +0.125]	0.772
REF-DOA SD 8 ms (oracle)	0.642	+0.183 [+0.141, +0.221]	<0.001	+0.134 [+0.093, +0.175]	<0.001
REF-DOA DS time-domain (oracle)	0.575	+0.116 [+0.082, +0.144]	<0.001	+0.067 [+0.031, +0.102]	<0.001

5. Breakdown by Target_of_Speech

Word-correct, Addressed to the wearer (n = 85)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.274			-0.055 [-0.105, +0.001]	

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Ear-canal mics mean ch5/6 (reference only)	0.256	-0.018 [-0.092, +0.058]		-0.073 [-0.139, -0.008]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.329	+0.055 [+0.001, +0.108]			
Head-steered DS, time-domain (ATF delays)	0.313	+0.039 [-0.004, +0.086]	0.380	-0.016 [-0.066, +0.032]	1.000
Head-steered superdirective 8 ms	0.346	+0.072 [+0.021, +0.136]	0.063	+0.017 [-0.037, +0.089]	1.000
Head-steered superdirective 4 ms	0.328	+0.054 [-0.003, +0.115]	0.336	-0.001 [-0.050, +0.055]	1.000
Tracked DS time-domain	0.262	-0.012 [-0.065, +0.033]	0.828	-0.067 [-0.117, -0.021]	0.027
Tracked SD 8 ms	0.241	-0.033 [-0.085, +0.015]	0.592	-0.088 [-0.135, -0.041]	0.011
Tracked SD 4 ms	0.251	-0.024 [-0.074, +0.017]	0.828	-0.078 [-0.123, -0.040]	0.011
Head-prior hybrid DS time-domain	0.295	+0.021 [-0.027, +0.069]	0.828	-0.033 [-0.081, +0.013]	1.000
Head-prior hybrid SD 8 ms	0.332	+0.058 [+0.010, +0.113]	0.176	+0.003 [-0.044, +0.060]	1.000
Head-prior hybrid SD 4 ms	0.328	+0.054 [+0.001, +0.113]	0.329	-0.001 [-0.050, +0.060]	1.000
REF-DOA SD 8 ms (oracle)	0.409	+0.135 [+0.082, +0.193]	<0.001	+0.080 [+0.022, +0.143]	0.056
REF-DOA DS time-domain (oracle)	0.383	+0.108 [+0.060, +0.161]	<0.001	+0.054 [+0.013, +0.106]	0.077

Word-correct, Addressed to the group (n = 125)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.378			-0.052 [-0.086, -0.013]	
Ear-canal mics mean ch5/6 (reference only)	0.291	-0.088 [-0.134, -0.044]		-0.140 [-0.183, -0.096]	

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.430	+0.052 [+0.016, +0.085]			
Head-steered DS, time-domain (ATF delays)	0.348	-0.030 [-0.064, +0.005]	0.165	-0.082 [-0.125, -0.038]	<0.001
Head-steered superdirective 8 ms	0.426	+0.047 [+0.009, +0.089]	0.070	-0.005 [-0.053, +0.044]	1.000
Head-steered superdirective 4 ms	0.416	+0.037 [-0.000, +0.075]	0.165	-0.015 [-0.061, +0.037]	1.000
Tracked DS time-domain	0.327	-0.051 [-0.080, -0.019]	<0.001	-0.104 [-0.133, -0.073]	<0.001
Tracked SD 8 ms	0.325	-0.054 [-0.088, -0.013]	0.070	-0.106 [-0.147, -0.063]	<0.001
Tracked SD 4 ms	0.316	-0.063 [-0.099, -0.027]	0.016	-0.115 [-0.153, -0.075]	<0.001
Head-prior hybrid DS time-domain	0.351	-0.027 [-0.064, +0.012]	0.172	-0.079 [-0.123, -0.031]	0.006
Head-prior hybrid SD 8 ms	0.423	+0.045 [+0.000, +0.093]	0.165	-0.007 [-0.057, +0.050]	1.000
Head-prior hybrid SD 4 ms	0.424	+0.046 [+0.001, +0.095]	0.165	-0.006 [-0.063, +0.051]	1.000
REF-DOA SD 8 ms (oracle)	0.520	+0.142 [+0.096, +0.192]	<0.001	+0.089 [+0.047, +0.137]	<0.001
REF-DOA DS time-domain (oracle)	0.451	+0.073 [+0.037, +0.106]	0.009	+0.020 [-0.014, +0.056]	1.000

Word-correct, Target: other (n = 123)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.265			-0.085 [-0.125, -0.048]	
Ear-canal mics mean ch5/6 (reference only)	0.193	-0.072 [-0.123, -0.024]		-0.157 [-0.213, -0.106]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.350	+0.085 [+0.046, +0.123]			

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Head-steered DS, time-domain (ATF delays)	0.245	-0.020 [-0.059, +0.016]	1.000	-0.104 [-0.152, -0.056]	<0.001
Head-steered superdirective 8 ms	0.259	-0.006 [-0.050, +0.043]	1.000	-0.091 [-0.148, -0.025]	0.024
Head-steered superdirective 4 ms	0.270	+0.005 [-0.041, +0.055]	1.000	-0.079 [-0.134, -0.021]	0.024
Tracked DS time-domain	0.229	-0.037 [-0.071, +0.001]	0.366	-0.121 [-0.159, -0.075]	<0.001
Tracked SD 8 ms	0.218	-0.047 [-0.092, -0.005]	0.264	-0.132 [-0.175, -0.088]	<0.001
Tracked SD 4 ms	0.220	-0.045 [-0.094, +0.000]	0.350	-0.130 [-0.177, -0.085]	<0.001
Head-prior hybrid DS time-domain	0.227	-0.039 [-0.076, -0.002]	0.207	-0.123 [-0.174, -0.073]	<0.001
Head-prior hybrid SD 8 ms	0.271	+0.005 [-0.051, +0.064]	1.000	-0.079 [-0.136, -0.013]	0.024
Head-prior hybrid SD 4 ms	0.256	-0.009 [-0.062, +0.045]	1.000	-0.094 [-0.149, -0.027]	0.024
REF-DOA SD 8 ms (oracle)	0.458	+0.193 [+0.146, +0.245]	<0.001	+0.108 [+0.063, +0.161]	<0.001
REF-DOA DS time-domain (oracle)	0.404	+0.139 [+0.100, +0.180]	<0.001	+0.054 [+0.023, +0.087]	<0.001

Word-correct, Target: none (n = 101)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.452			-0.067 [-0.114, -0.024]	
Ear-canal mics mean ch5/6 (reference only)	0.335	-0.117 [-0.193, -0.047]		-0.184 [-0.280, -0.112]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.519	+0.067 [+0.024, +0.115]			
Head-steered DS, time-domain (ATF delays)	0.497	+0.045 [-0.076, +0.127]	0.846	-0.022 [-0.168, +0.088]	1.000
Head-steered superdirective 8 ms	0.625	+0.174 [+0.079, +0.251]	0.010	+0.107 [-0.009, +0.210]	0.365

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Head-steered superdirective 4 ms	0.618	+0.166 [+0.065, +0.250]	0.016	+0.099 [-0.022, +0.204]	0.365
Tracked DS time-domain	0.424	-0.028 [-0.061, +0.010]	0.372	-0.095 [-0.156, -0.047]	<0.001
Tracked SD 8 ms	0.329	-0.122 [-0.182, -0.065]	<0.001	-0.189 [-0.284, -0.094]	<0.001
Tracked SD 4 ms	0.343	-0.108 [-0.174, -0.054]	<0.001	-0.176 [-0.289, -0.087]	<0.001
Head-prior hybrid DS time-domain	0.500	+0.049 [-0.077, +0.152]	0.846	-0.019 [-0.168, +0.106]	1.000
Head-prior hybrid SD 8 ms	0.642	+0.190 [+0.075, +0.272]	0.007	+0.123 [-0.004, +0.229]	0.276
Head-prior hybrid SD 4 ms	0.618	+0.167 [+0.068, +0.246]	0.007	+0.100 [-0.016, +0.198]	0.365
REF-DOA SD 8 ms (oracle)	0.669	+0.217 [+0.168, +0.264]	<0.001	+0.150 [+0.084, +0.217]	<0.001
REF-DOA DS time-domain (oracle)	0.592	+0.141 [+0.112, +0.166]	<0.001	+0.073 [+0.024, +0.116]	0.014

6. Time since turn onset

Word-correct, < 0.5 s since turn onset (n = 371)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.320			-0.073 [-0.095, -0.049]	
Ear-canal mics mean ch5/6 (reference only)	0.258	-0.062 [-0.092, -0.031]		-0.135 [-0.172, -0.103]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.394	+0.073 [+0.050, +0.094]			
Head-steered DS, time-domain (ATF delays)	0.318	-0.002 [-0.031, +0.024]	0.904	-0.075 [-0.105, -0.042]	<0.001
Head-steered superdirective 8 ms	0.380	+0.059 [+0.025, +0.095]	<0.001	-0.014 [-0.051, +0.026]	1.000
Head-steered superdirective 4 ms	0.372	+0.051 [+0.018, +0.086]	0.006	-0.022 [-0.058, +0.018]	1.000

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Tracked DS time-domain	0.289	-0.031 [-0.054, -0.009]	0.024	-0.105 [-0.128, -0.080]	<0.001
Tracked SD 8 ms	0.267	-0.053 [-0.078, -0.027]	<0.001	-0.126 [-0.153, -0.097]	<0.001
Tracked SD 4 ms	0.263	-0.057 [-0.086, -0.030]	<0.001	-0.130 [-0.161, -0.100]	<0.001
Head-prior hybrid DS time-domain	0.309	-0.012 [-0.036, +0.015]	0.746	-0.085 [-0.114, -0.052]	<0.001
Head-prior hybrid SD 8 ms	0.381	+0.061 [+0.024, +0.100]	0.006	-0.012 [-0.052, +0.031]	1.000
Head-prior hybrid SD 4 ms	0.371	+0.051 [+0.016, +0.088]	0.006	-0.023 [-0.062, +0.019]	1.000
REF-DOA SD 8 ms (oracle)	0.497	+0.177 [+0.147, +0.206]	<0.001	+0.104 [+0.073, +0.134]	<0.001
REF-DOA DS time-domain (oracle)	0.431	+0.111 [+0.089, +0.131]	<0.001	+0.038 [+0.014, +0.061]	0.005

Word-correct, 0.5-2 s since turn onset (n = 11)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.299			-0.133 [-0.347, +0.041]	
Ear-canal mics mean ch5/6 (reference only)	0.330	+0.030 [-0.103, +0.199]		-0.102 [-0.219, +0.017]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.432	+0.133 [-0.053, +0.346]			
Head-steered DS, time-domain (ATF delays)	0.481	+0.182 [-0.002, +0.399]	0.470	+0.049 [-0.020, +0.136]	1.000
Head-steered superdirective 8 ms	0.443	+0.144 [-0.042, +0.367]	0.900	+0.011 [-0.101, +0.123]	1.000
Head-steered superdirective 4 ms	0.469	+0.170 [-0.019, +0.380]	0.846	+0.037 [-0.060, +0.141]	1.000
Tracked DS time-domain	0.301	+0.002 [-0.034, +0.039]	0.910	-0.131 [-0.340, +0.048]	1.000
Tracked SD 8 ms	0.214	-0.085 [-0.196, +0.019]	0.846	-0.218 [-0.401, -0.082]	<0.001
Tracked SD 4 ms	0.226	-0.074 [-0.195, +0.031]	0.900	-0.207 [-0.401, -0.077]	<0.001

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Head-prior hybrid DS time-domain	0.407	+0.107 [-0.062, +0.322]	0.900	-0.026 [-0.117, +0.080]	1.000
Head-prior hybrid SD 8 ms	0.435	+0.135 [-0.045, +0.359]	0.900	+0.002 [-0.107, +0.113]	1.000
Head-prior hybrid SD 4 ms	0.435	+0.135 [-0.056, +0.359]	0.900	+0.002 [-0.105, +0.103]	1.000
REF-DOA SD 8 ms (oracle)	0.470	+0.171 [-0.035, +0.394]	0.846	+0.038 [-0.053, +0.129]	1.000
REF-DOA DS time-domain (oracle)	0.528	+0.229 [+0.085, +0.418]	<0.001	+0.096 [+0.001, +0.199]	0.513

Word-correct, > 2 s since turn onset (n = 52)

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Unprocessed frame mic (frozen best of ch1-4)	0.513			+0.005 [-0.063, +0.057]	
Ear-canal mics mean ch5/6 (reference only)	0.312	-0.202 [-0.288, -0.105]		-0.197 [-0.332, -0.065]	
R5 DS, blind whole-utt. dir., 32 ms STFT (reference baseline)	0.509	-0.005 [-0.059, +0.069]			
Head-steered DS, time-domain (ATF delays)	0.521	+0.008 [-0.132, +0.113]	1.000	+0.013 [-0.195, +0.160]	1.000
Head-steered superdirective 8 ms	0.616	+0.102 [-0.005, +0.190]	0.415	+0.107 [-0.047, +0.231]	1.000
Head-steered superdirective 4 ms	0.624	+0.110 [-0.002, +0.189]	0.308	+0.115 [-0.043, +0.228]	1.000
Tracked DS time-domain	0.453	-0.060 [-0.129, +0.024]	0.525	-0.055 [-0.133, +0.040]	1.000
Tracked SD 8 ms	0.378	-0.136 [-0.215, -0.065]	<0.001	-0.131 [-0.284, -0.026]	0.010
Tracked SD 4 ms	0.431	-0.083 [-0.157, -0.004]	0.240	-0.078 [-0.214, +0.021]	1.000
Head-prior hybrid DS time-domain	0.548	+0.035 [-0.083, +0.133]	1.000	+0.040 [-0.156, +0.193]	1.000
Head-prior hybrid SD 8 ms	0.635	+0.121 [+0.003, +0.205]	0.308	+0.126 [-0.025, +0.246]	0.968

method	word-correct	Δ vs unproc_front [95% CI]	Holm p	Δ vs R5 DS [95% CI]	Holm p
Head-prior hybrid SD 4 ms	0.625	+0.111 [-0.012, +0.195]	0.415	+0.116 [-0.044, +0.236]	1.000
REF-DOA SD 8 ms (oracle)	0.656	+0.143 [+0.091, +0.173]	<0.001	+0.148 [+0.074, +0.209]	<0.001
REF-DOA DS time-domain (oracle)	0.627	+0.114 [+0.039, +0.169]	0.063	+0.119 [+0.029, +0.198]	0.117

Few items fall after 0.5 s (n = 63), so the later bins are wide and mostly uninformative; most test items are short turn starts.

7. Head vs tracker steering accuracy

Share of 50 ms frames with the steering direction within X° of the true talker, by time since the target's turn started (all test and val turns, n = 1565 turns; results/r7/onset_r7.csv, head_onset.csv):

steering	0-0.25 s, ≤30°	0-0.25 s, ≤10°	> 4 s, ≤30°	> 4 s, ≤10°
head azimuth, all	54%	21%	75%	27%
head azimuth, to wearer	71%	32%	89%	31%
head azimuth, overlapped	46%	15%	69%	22%
tracker (ATF), all	35%	14%	71%	49%
tracker (ATF), to wearer	56%	21%	78%	50%
best of head or tracker (oracle pick)	59%	28%	84%	57%

Head beats tracker for the first seconds and especially at onsets; the tracker is more precise once it has locked on. The "best of" row shows a complementary signal, but my hybrid rule (head prior plus a confidence-gated tracker) realised +0.004 over head alone, not the gap implied.

Figure: results/r7/figs/head_onset.png.

8. Steering error vs gain

Per item, median steering error vs word-correct gain over front mic (head SD 8 ms): 0-10° → +0.156 (n = 152); 10-20° → +0.119 (82); 20-45° → +0.010 (108); 45-90° → −0.069 (81). Hybrid SD 8 ms: +0.159 (0-10°, n = 165), +0.100, +0.015, −0.082. Spearman $\rho = -0.37$. Median head-angle off the talker is 4° for the best bin and 61° for the worst, so a head-centred beam helps only when the wearer is facing the talker, which happens in about 35% of items.

9. Critique and limitations

- **Proxy metric.** Whisper base.en word-correct, not human listeners. Truncation of long items by the ASR is not fixed.
- **Reference built from poses.** Talker position = head-pose centre (no mouth offset); alignment of 20 test items was suspect (results with and without them agree).
- **Only a modest tie with R5 DS.** The 8 ms head-steered beam does not beat the non-causal blind direction. The oracle shows +0.107 over R5 DS is available, so direction remains the bottleneck.

- **Small test set.** Two sessions, so the cluster bootstrap CIs are wide. Subgroup gains (overlap, wearer) do not survive Holm correction.
- **Hybrid adds nothing.** The tracker's confidence rarely crosses the threshold in the short items; it may help in longer turns.
- **Geometry.** 4 mics, 12 cm aperture, nearly horizontal; low-frequency gain is small (WNG floor −6 dB).
- **Not tested:** human listening, moving talkers beyond the recorded sessions, closed-loop open-ear leakage.

10. Next steps

1. Learn a talker-selection prior from head pose plus voice activity (who is speaking to the wearer) rather than a fixed look direction.
2. Try gaze-like cues (head velocity, turn-taking) in the hybrid to recover the head/tracker complementarity.
3. A human listening test (12+ people) with level-matched head-steered vs front vs R5 DS.
4. Re-bench the real-time demo (idle load) and swap its mode (b) to head-steered SD 8 ms.

11. Files

results/r7/summary_r7.md (all tables), results/r7/{head_onset,onset_r7,steer_test,asr_wer_test,timing_r7_summary}.csv, results/r7/frozen_r7.json, results/r7/figs/head_onset.png, selab/{run_r7,analyze_r7,timing_r7,summary_r7,bf_r7}.py.
No EasyCom audio is included.